MODULE 16 · DOMAIN 2 - RELIABILITY AND BUSINESS CONTINUITY

Backup and Restore

AWS Backup, snapshots, point-in-time recovery, Object Lock and Vault Lock - and proving any of it works.

DOMAIN 2 of 5 EXAM WEIGHT 22% MODULE 16 of 35

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A backup you have never restored is a **hypothesis**. The exam tests the restore path at least as often as the backup path.

Native backup, service by service

SERVICE	NATIVE MECHANISM
Amazon EBS	Snapshots, incremental. Automated by Data Lifecycle Manager
Amazon RDS	Automated backups with PITR, plus manual snapshots
Amazon Aurora	Continuous backup to S3, plus Backtrack on MySQL and cloning
Amazon DynamoDB	Point-in-time recovery for 35 days, plus on-demand backups
Amazon S3	Versioning, replication and Object Lock. Not snapshots
Amazon EFS and FSx	AWS Backup. EFS has no native snapshot of its own

ON THE EXAM

Every one of these can also be driven centrally by AWS Backup, which is usually the answer once more than one service is involved.

RDS automated backups or manual snapshots?

AUTOMATED BACKUPS

Continuous, and temporary

- Retention 0 to 35 days; 0 disables them
- Enable point-in-time recovery within that window
- Taken during the backup window you configure
- Deleted with the instance unless you retain them

MANUAL SNAPSHOTS

A point in time, kept

- Taken on demand, kept until you delete them
- Survive deletion of the DB instance
- Can be copied across Regions and accounts
- No point-in-time recovery from a snapshot alone

THE TIEBREAKER

Automated backups die with the instance. If an accidental deletion must be survivable, a manual snapshot or an AWS Backup vault copy is required.

What an RDS restore actually gives you

- **A restore always creates a NEW DB instance**, with a **new endpoint**. You cannot restore in place.
- So a restore is followed by **repointing the application**, which is part of your RTO whether you planned for it or not.
- **Point-in-time recovery** restores to any second in the retention window, typically to within about 5 minutes of now.
- The new instance uses the **default parameter and security groups** unless you specify otherwise. That surprises people mid-incident.

> ON THE EXAM
>
> Practise the repoint, not just the restore. A recovered database nobody can reach has not recovered anything.

Aurora: backtrack and cloning

- **Backtrack** rewinds an Aurora MySQL cluster **in place**, up to 72 hours, in minutes. No new instance and no new endpoint.
- **MySQL-compatible only**, must be enabled in advance, and it is **not a backup**: it cannot recover a deleted cluster.
- **Cloning** creates a copy-on-write clone of the cluster volume in minutes, paying only for changed pages.
- Clones are how you get a realistic test environment, or a safe place to try a destructive migration.

> ON THE EXAM
>
> Backtrack is the only mechanism here that recovers in place. That is what makes it fast, and also why it cannot survive the cluster being deleted.

EBS snapshots

- **Incremental**: each snapshot stores only blocks changed since the last one, but any single snapshot can be deleted safely.
- **Data Lifecycle Manager** schedules them by tag, with retention, cross-Region copy and cross-account sharing.
- **Fast Snapshot Restore** removes first-touch latency on a restored volume. Charged per snapshot per AZ per hour.
- **Snapshot Archive** is cheaper for long retention, with a **90-day minimum** and a restore that can take up to 72 hours.

> ON THE EXAM
>
> A restored volume is normally lazily loaded from S3, so the first read of each block is slow. That is what Fast Snapshot Restore removes.

DynamoDB backup

- **Point-in-time recovery** covers the last **35 days**, to any second. Enabled per table, off by default.
- **On-demand backups** are full backups kept until deleted, with no effect on table performance or capacity.
- **Both restore to a NEW table.** The same lesson as RDS: recovery includes repointing the application.
- **Export to S3** is a third path, for analytics rather than recovery. It consumes no read capacity.

ON THE EXAM

A restored DynamoDB table does not carry over auto scaling settings, TTL configuration, or Streams. Check those after any restore.

S3 versioning: what it does and does not protect

- **Protects against overwrite and delete.** A delete creates a **delete marker**; the previous version is still there.
- **MFA delete** requires a hardware or virtual MFA token to delete a version or disable versioning.
- It does **not** protect against someone with permission deleting **every version**, which is exactly what ransomware does.
- Versioning **cannot be switched off** once enabled, only suspended, and old versions keep costing storage until a lifecycle rule removes them.

> ON THE EXAM
>
> Versioning is an undo button. Object Lock is a padlock. A scenario about a malicious insider or ransomware needs the padlock.

S3 Object Lock

- Write once, read many. **Requires versioning**, and is normally enabled at bucket creation.
- **Governance mode**: protected, but a user with the **BypassGovernanceRetention** permission can override it.
- **Compliance mode**: nobody can shorten or remove the retention, **including the root user**, until it expires.
- **Legal hold** has no expiry. It blocks deletion indefinitely until someone with permission removes it.

> ON THE EXAM
>
> "Including administrators" or "including the root user" in a requirement always means compliance mode. Governance mode is designed to be overridable.

AWS Backup: the central service

- A **backup plan** contains **rules**: schedule, backup window, lifecycle to cold storage, retention, and copy destinations.
- A **resource selection** chooses what the plan covers, usually **by tag**, so new resources are protected automatically.
- Backups land in a **vault**, which is the unit of encryption, access policy and locking.
- **Organization backup policies** push plans from the management account so member accounts cannot opt out.

> ON THE EXAM
>
> Tag-based selection is the point. "New resources must be backed up without anyone remembering" is a tag-driven AWS Backup plan.

Copies: across Regions and accounts

- **Cross-Region copy** is a rule in the plan. It is how backups survive a Regional event.
- **Cross-account copy** is the ransomware answer: backups live in an account the compromised one cannot reach.
- A copy in the **same account** protects against deletion of the original resource, not against the account being compromised.
- Encrypted copies need the destination vault's **KMS key policy** to permit the copy. This is the usual failure.

> ON THE EXAM
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> Ask what you are defending against. A mistake needs another copy. An attacker with your credentials needs another account.

Vault Lock: governance or compliance?

GOVERNANCE MODE

Guard rail

- Enforces retention against ordinary users
- Privileged users can remove the lock
- Right for preventing mistakes
- Reversible, so safe to adopt early

COMPLIANCE MODE

One-way door

- After the grace period, nobody can remove it
- Not the root user, not AWS Support
- Retention cannot be shortened, backups cannot be deleted
- The only exit is closing the AWS account

THE TIEBREAKER

Compliance mode has a grace period, minimum three days, during which it can still be cancelled. After that it is permanent. Test in governance mode first.

Logically air-gapped vaults

- A vault type that is **automatically locked in compliance mode** and encrypted, generally available since August 2024.
- Can be **shared with other accounts through AWS RAM**, including accounts outside your organisation.
- Shared accounts can **restore directly from it**, with no copy step first. That is what compresses recovery time.
- Since **November 2025** it can be the **primary backup target**, not only a copy destination.

> ON THE EXAM
>
> The direct restore is the point. Traditional isolated backups had to be copied somewhere usable first, and that copy was pure RTO.

New: scanning backups before you restore them

- **GuardDuty Malware Protection for AWS Backup**, generally available **November 2025**.
- Scans backup data with multiple detection engines. **No agent and no extra security software** on the workload.
- Covers **EC2 instance backups, EBS snapshots and S3 backups**.
- It addresses the failure nobody plans for: **restoring the malware along with the data**.

> ON THE EXAM
>
> Ransomware often sits dormant for weeks before triggering, so the backups you trust may already contain it. Scanning before restore is the control for that.

Proving it works

- **Restore testing** runs scheduled, automated restore jobs and validates them, then cleans up the restored resources.
- It produces evidence of **recovery time**, which is the only honest way to know your RTO.
- **Backup Audit Manager** checks controls: is everything tagged for backup actually backed up, and to the required retention?
- It reports on **coverage gaps**, which is the failure mode where a resource was simply never included in any plan.

> ON THE EXAM
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> "Demonstrate to an auditor that backups exist, meet retention, and can be restored" is Backup Audit Manager plus restore testing, not a spreadsheet.

What restore actually produces

- **RDS and DynamoDB**: a new instance or table, with a new endpoint or name. Repointing is your job.
- **EBS**: a new volume, which must then be attached, and possibly mounted, before anything sees it.
- **Aurora Backtrack**: the same cluster, rewound in place. The only true in-place recovery here.
- **S3 versioning**: delete the delete marker and the object is back. The closest thing to a genuine undo.

> ON THE EXAM
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> Rank recovery mechanisms by how much work happens after the restore finishes. That work is invisible in the backup plan and very visible during an incident.

> An engineer deletes a production RDS for MySQL instance by mistake. Automated backups were enabled with 7-day retention, and the 'retain automated backups' option was not selected on deletion. There are no manual snapshots.
>
> The team asks how to recover the database to five minutes before the deletion.

What is the situation?

A Restore to the required point in time from the automated backups

B The automated backups were deleted with the instance, so there is nothing to restore from

C Use Aurora Backtrack to rewind past the deletion

D Restore from the final snapshot RDS always takes automatically on deletion

◆ PAUSE AND DECIDE

What is the situation?

A Restore from automated backups

There are none left. Automated backups are removed with the instance unless retention was explicitly requested at deletion.

B Nothing to restore from

Uncomfortable and correct. This is precisely why manual snapshots or an AWS Backup vault copy exist as a separate control.

C Aurora Backtrack

MySQL-compatible Aurora only, not RDS for MySQL. It also cannot rewind a cluster that no longer exists.

D The automatic final snapshot

A final snapshot is offered at deletion, not taken automatically. The scenario says it was not selected.

THE SIGNAL

Some questions have an unpleasant correct answer. Recognising that a recovery is impossible is a real skill, and the exam does test it.

A regulator covers seven years of financial records in S3. Nobody in the company may delete or alter them for the full retention period. That includes administrators and the root user.

The bucket currently has versioning enabled and a bucket policy denying deletes to everyone except a small admin group.

What should the company implement?

A S3 Object Lock in compliance mode with a seven-year retention period

B S3 Object Lock in governance mode with a seven-year retention period

C Enable MFA delete on the bucket

D Cross-Region replication to a bucket in another account

◆ PAUSE AND DECIDE

What should the company implement?

A Object Lock, compliance mode

The only control here that the root user cannot override. Retention cannot be shortened and objects cannot be deleted until it expires.

B Object Lock, governance mode

Designed to be overridable by a user holding BypassGovernanceRetention. The requirement says nobody, which rules that out.

C MFA delete

Raises the bar and does not remove the capability. An administrator with the MFA device can still delete.

D Replication to another account

Improves durability and isolation. It does not make anything immutable, and the copies are deletable too.

THE SIGNAL

Same discriminator as the CloudTrail module: "including the root user" eliminates every control that a privileged user can undo.

A company is compromised by ransomware. The attacker had administrator credentials in the production account for several days before triggering, and deleted every EBS snapshot and RDS snapshot in that account. Backups were being copied to a second vault in the same account.

The company wants a design that survives this next time, and wants recovery to begin without a lengthy copy step.

What should the company implement?

A. A logically air-gapped vault in a separate account, shared for direct restore via AWS RAM

B. A second backup vault in the production account with Vault Lock in governance mode

C. Cross-Region copy to another Region in the same account

D. Increase the retention period on the existing backup plan

◆ PAUSE AND DECIDE

What should the company implement?

A Air-gapped vault in a separate account

Compliance-locked by default, in an account the attacker had no credentials for, and RAM sharing allows restore directly without copying first.

B Governance-mode vault, same account

Fails twice: an administrator can remove a governance lock, and it is in the account that was compromised.

C Cross-Region, same account

Another Region is not another blast radius when the credentials are account-wide. The attacker can delete it too.

D Longer retention

Keeps deleted backups for longer, which helps nobody once they have been deleted.

THE SIGNAL

Two clauses drive it: the attacker had administrator credentials, and recovery must not wait for a copy. Only the air-gapped vault answers both.

A team restores an RDS database from a snapshot during a disaster recovery test. The restore itself completes in about 25 minutes. The team records an RTO of 25 minutes.

During a real incident three months later, the total outage lasts nearly two hours despite the restore again taking 25 minutes.

What most likely accounts for the difference?

A The snapshot was in cold storage and had to be retrieved

B The restore creates a new instance with a new endpoint, and repointing the application, security groups and parameter groups was never measured

C Fast Snapshot Restore was not enabled

D Point-in-time recovery is slower than snapshot restore

◆ PAUSE AND DECIDE

What most likely accounts for the difference?

A **Cold storage retrieval**

Would have made the restore itself slower. The scenario states the restore took the same 25 minutes both times.

B **Everything after the restore was never measured**

The restore produces a new endpoint. Repointing, fixing security groups and parameter groups is real work that the test skipped.

C **Fast Snapshot Restore**

An EBS feature affecting volume first-touch latency. It does not apply to an RDS snapshot restore.

D **PITR is slower**

The restore duration is stated as identical in both cases, so the mechanism's speed is not the variable.

THE SIGNAL

The scenario fixes the restore time at 25 minutes twice, deliberately. Whatever explains the difference has to be outside the restore.

> An operations team must ensure that every EBS volume and RDS database created by any of 30 accounts is backed up daily, retained for 90 days, and copied to a second Region, with no manual step when new resources are created.
>
> They must also be able to show an auditor which resources are not covered.

Which combination should they implement?

A. An organization backup policy with tag-based resource selection and a cross-Region copy rule, plus Backup Audit Manager

B. Data Lifecycle Manager policies in each account, plus RDS automated backups

C. A Lambda function on an EventBridge schedule that snapshots every volume and database

D. AWS Backup plans created individually in each of the 30 accounts

◆ PAUSE AND DECIDE

Which combination should they implement?

A Organization policy plus Audit Manager

One policy pushed to all 30 accounts, tag selection covers resources created later, and Audit Manager reports the coverage gaps the auditor wants.

B DLM plus RDS automated backups

DLM covers EBS only, RDS automated backups do not do cross-Region copy this way, and both need configuring in every account.

C Lambda on a schedule

Rebuilds AWS Backup by hand across 30 accounts, with no coverage reporting and a function to maintain forever.

D Plans in each account

Works and drifts. Thirty independently maintained plans is exactly the divergence an organization policy exists to prevent.

THE SIGNAL

Three requirements: automatic coverage of new resources, thirty accounts, and evidence of gaps. Tag selection, organization policy, Audit Manager.

The five things to carry forward

- **Automated backups die with the instance.** Manual snapshots and vault copies survive it.
- **Restore creates something new.** Repointing is part of your RTO, and Aurora Backtrack is the only in-place exception.
- **Versioning is undo; Object Lock is a padlock.** "Including the root user" means compliance mode.
- **Another Region for a Regional event; another account for a compromised account.**
- **Restore testing measures RTO; Audit Manager finds coverage gaps.** GuardDuty scans backups before you restore malware.

> ON THE EXAM
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> Next: disaster recovery proper - RPO, RTO, the four strategies, and Application Recovery Controller.